Do not write outside the box

0 5 During electrolysis ions are discharged at the electrodes to produce elements.

A student investigates the electrolysis of sodium chloride.

0 5 . 1 Why does solid sodium chloride **not** conduct electricity?

[1 mark]

_____

_____

0 5 . 2 Sodium chloride solution conducts electricity.

Complete the sentence.

[1 mark]

Sodium chloride **also** conducts electricity when _____ .

Question 5 continues on the next page

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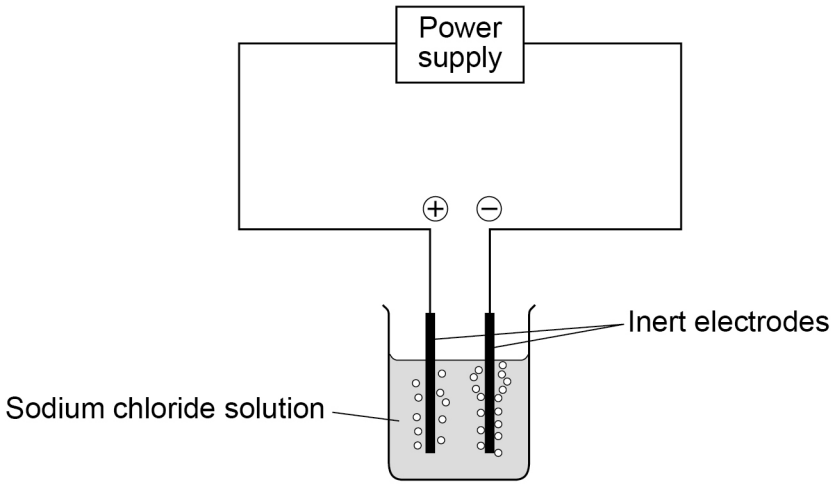
Figure 5 shows the apparatus for the electrolysis of sodium chloride solution.

Figure 5



0 5 . 3 Suggest an element that could be used to make the inert electrodes.

[1 mark]

_____

0 5 . 4 Complete the half equation for the production of chlorine (Cl_2) at the positive electrode.

[2 marks]

____ Cl^- → _____ + _____

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0 5 . 5 Sodium chloride solution has a pH of 7

During the electrolysis of sodium chloride solution:

- hydrogen gas is produced at the negative electrode
- the pH of the solution increases.

Explain why.

[4 marks]

9

Turn over for the next question